

LSIS-AFS Competition — Requirements Traceability Matrix

This document maps every testable requirement to its LSIS specification source, acceptance criteria, and scoring weight. Use it as a checklist to track your progress and understand exactly what will be measured during evaluation.

How to Read This Matrix

Column	Meaning
ID	Unique requirement identifier
LSIS Ref	Section/table in LSIS-AFS Volume A
Requirement	What you must implement
Acceptance Criteria	How we verify it passes
Gateway	Which development milestone it belongs to
Points	Scoring weight (out of 100 total)

Minimum Viable Submission (Pass/Fail Gate)

Your submission is not scored unless ALL of the following are met:

#	Criterion	How We Test
MVS-1	Generate at least 12 PRN spreading codes correctly	Compare hex output against Annex3 for PRN 1-12
MVS-2	Encode a complete 12-second navigation frame	Verify 6000 symbols output with correct structure
MVS-3	Generate I/Q signal files	Verify file is readable and contains valid samples
MVS-4	Pass validation tests for implemented features	Run your own test suite, all must pass
MVS-5	Include setup/build instructions that work	We follow your README from a clean machine

Category 1: Correctness (40 points)

1.1 Spreading Code Generation (10 points)

ID	LSIS Ref	Requirement	Acceptance Criteria	Points
COR-1.1	LSIS-221, Appendix C, Annex3	Gold code generation (2046 chips, PRN 1-210)	All 210 codes match Annex3 hex file exactly	4
COR-1.2	LSIS-222, Appendix D, Annex3	Weil primary code generation (10230 chips, PRN 1-210)	All 210 codes match Annex3 hex file exactly	3
COR-1.3	LSIS-224, Appendix E, Annex3	Weil tertiary code generation (1500 chips, PRN 1-210)	All 210 codes match Annex3 hex file exactly	3

How we test: Generate all 210 PRN codes in hex format, byte-compare against Annex3 reference files.

1.2 Frame Encoding (10 points)

ID	LSIS Ref	Requirement	Acceptance Criteria	Points
COR-2.1	Section 2.4.2, Figure 6	BCH(51,8) encoding with polynomial 763 (octal)	Known input produces expected 52-symbol output	2
COR-2.2	Section 2.4.3, Annex1	LDPC rate 1/2 encoding for SB2 (1200 bits -> 2400 symbols)	Valid codeword ($H^*c^T = 0$), puncturing correct	3
COR-2.3	LSIS-FID0-467	CRC-24 with specified generator polynomial	CRC matches for known test vectors	2
COR-2.4	Section 2.4.3, Table 17	Block interleaver 60x98 (write row-wise, read column-wise)	Round-trip deinterleave recovers original	1
COR-2.5	Section 2.4, Figure 9	Frame assembly: SP(68) + SB1(52) + interleaved(5880) = 6000	Symbol count exactly 6000, structure correct	2

How we test: Provide known navigation data, verify encoded frame matches expected output bit-for-bit.

1.3 Signal Generation and Decoding (10 points)

ID	LSIS Ref	Requirement	Acceptance Criteria	Points
COR-3.1	Table 8	BPSK modulation: logic 0 -> +1.0, logic 1 -> -1.0	Sample values correct for known symbol sequence	2
COR-3.2	Table 7, LSIS-160	AFS-I spreading at 1.023 Mchip/s, 500 sym/s	Each symbol spread by 2046 chips	2
COR-3.3	Section 2.4.1, Table 12	Frame synchronization via 68-symbol sync pattern	Detect frame start in clean signal	3
COR-3.4	—	Round-trip: encode -> signal -> decode -> verify	Original navigation data recovered exactly	3

How we test: Generate signal from known data, decode it, verify bit-exact recovery. Also test with our reference encoder's output.

1.4 Specification Compliance (10 points)

ID	LSIS Ref	Requirement	Acceptance Criteria	Points
COR-4.1	Table 9	Code lengths: Gold=2046, Weil=10230, Tertiary=1500, Secondary=4	Exact lengths	2
COR-4.2	Section 2.4	Frame duration exactly 12 seconds	6000 symbols at 500 sym/s = 12s	2
COR-4.3	Table 7	Symbol rate 500 sym/s (AFS-I), chip rates per spec	Timing parameters correct	2

ID	LSIS Ref	Requirement	Acceptance Criteria	Points
COR-4.4	Tables 14, 18, 19, 20	Bit allocations for all subframes	Field positions and sizes match tables	2
COR-4.5	All "shall" statements	All mandatory requirements implemented	No deviations from mandatory spec	2

How we test: Automated compliance checker validates structural parameters against specification values.

Category 2: Performance (20 points)

2.1 Computational Efficiency (10 points)

ID	Requirement	Target	Measurement Method	Points
PERF-1.1	Code generation time	< 1 second per PRN	Average over all 210 PRNs	2
PERF-1.2	Frame encoding time	< 100 ms per frame	Average over 100 frames	3
PERF-1.3	Frame decoding time	< 1 second per frame	Average over 100 frames (clean signal)	3
PERF-1.4	Real-time capability	Process 12s frame in < 12s total	End-to-end encode+decode wall-clock	2

How we test: Run on standard benchmark hardware, average over multiple iterations, measure wall-clock time.

2.2 Error Correction Performance (10 points)

ID	Requirement	Target	Measurement Method	Points
PERF-2.1	Bit Error Rate at SNR = 0 dB	BER < 10^{-5}	1000 frames with AWGN	5
PERF-2.2	Bit Error Rate at SNR = 3 dB	BER < 10^{-6}	1000 frames with AWGN	3
PERF-2.3	Frame sync detection reliability	> 99% at SNR > 0 dB	1000 frames with AWGN	2

How we test: Add calibrated AWGN to generated signals, process through your decoder, measure error rates.

Category 3: Completeness (20 points)

3.1 Gateway Delivery (10 points)

ID	Gateway	Deliverable	Verification	Points
COMP-1.1	1	Spreading code generation (all types)	Codes generated and validated	1.5

ID	Gateway	Deliverable	Verification	Points
COMP-1.2	2	FEC encoding infrastructure (BCH, LDPC, CRC, interleaver)	Encoders produce valid output	1.5
COMP-1.3	3	Navigation message framing	Complete 6000-symbol frames generated	1.5
COMP-1.4	4	Baseband I/Q signal generation	Valid I/Q files produced	1.5
COMP-1.5	5	Frame sync and decoding	Frames decoded from signal	1
COMP-1.6	6	Message parsing	Navigation data extracted	1
COMP-1.7	7	Integration and end-to-end testing	Round-trip demonstrated	1
COMP-1.8	8	Documentation and examples	Usable by third party	1

3.2 Test Coverage (5 points)

ID	Requirement	Target	Points
COMP-2.1	Unit tests for all major components	Tests exist and pass	2
COMP-2.2	Integration tests for end-to-end flow	At least one round-trip test	1
COMP-2.3	Code coverage	> 90% of source lines	1
COMP-2.4	All tests pass	Zero failures in test suite	1

3.3 PRN Code Coverage (5 points)

ID	Requirement	Target	Points
COMP-3.1	Interim test codes (Table 11)	All 12 codes correct	3
COMP-3.2	Extended PRN coverage	At least 100 PRNs correct	1
COMP-3.3	Full PRN coverage	All 210 PRNs correct	1

Category 4: Code Quality (10 points)

ID	Requirement	What We Look For	Points
QUAL-1.1	Code readability	Clear naming, logical organization, useful comments	3
QUAL-1.2	Documentation	README, API docs, design decisions, working examples	4
QUAL-1.3	Software engineering	Modular design, error handling, reproducible builds	3

Category 5: Innovation and Extras (10 points)

ID	Requirement	Examples	Points
INNOV-1.1	Performance optimizations	SIMD, parallelism, algorithmic improvements	0-5
INNOV-1.2	Additional features	Visualization, advanced decoders, extra message types	0-3
INNOV-1.3	Novel approaches	Creative solutions, unique architecture	0-2

LSIS Specification Cross-Reference

The following table maps key LSIS requirement IDs to implementation areas:

LSIS Req ID	Description	Implementation Area
LSIS-160	Data symbol alignment with primary code	Signal generation
LSIS-170	Secondary code alignment with primary code	Tiered code assembly
LSIS-171	Tertiary code alignment with secondary code	Tiered code assembly
LSIS-172	Coherent tiered code generation	Tiered code assembly
LSIS-220	Tertiary code sync with frame start	Signal generation
LSIS-221	AFS-I primary code definition (Gold, 2046)	Code generation
LSIS-222	AFS-Q primary code definition (Weil, 10230)	Code generation
LSIS-223	AFS-Q secondary code definition (4 chips)	Code generation
LSIS-224	AFS-Q tertiary code definition (Weil, 1500)	Code generation
LSIS-260	Code assignment per LNSP node (Table 11)	Code database
LSIS-FID0-465	CRC guards all SB2/SB3/SB4 data	CRC-24
LSIS-FID0-467	CRC-24 generator polynomial	CRC-24

Data Files Required

File	Source	Used For
GoldCode2046hex210prns.txt	Annex3	Gold code validation
l1cp_hex210prns.txt	Annex3	Weil primary code validation
Weil1500hex210prns.txt	Annex3	Weil tertiary code validation
LDPC matrices (CSV)	Annex1	LDPC encoding/decoding
appendix_c_gold_params.csv	spec_tables	Gold code parameters
appendix_d_weil_primary_params.csv	spec_tables	Weil primary parameters
appendix_e_weil_tertiary_params.csv	spec_tables	Weil tertiary parameters

Scoring Summary

Category	Points	What Matters Most
Correctness	40	Bit-exact match with references, spec compliance
Performance	20	Speed targets met, BER curves acceptable
Completeness	20	All gateways delivered, tests pass, PRN coverage
Code Quality	10	Readable, documented, well-engineered
Innovation	10	Goes beyond minimum, creative solutions
Total	100	

Tiebreaker Rules

If two submissions have the same total score: 1. Higher Correctness score wins 2. If still tied: higher Performance score wins 3. If still tied: earlier submission timestamp wins